MULTIMODAL AI HACKATHON 2026 | TRACK B

HimaRoad Flood Copilot

A situation-report copilot fed by our own flood segmentation model

Track B: Mapping Flood Damage from Space

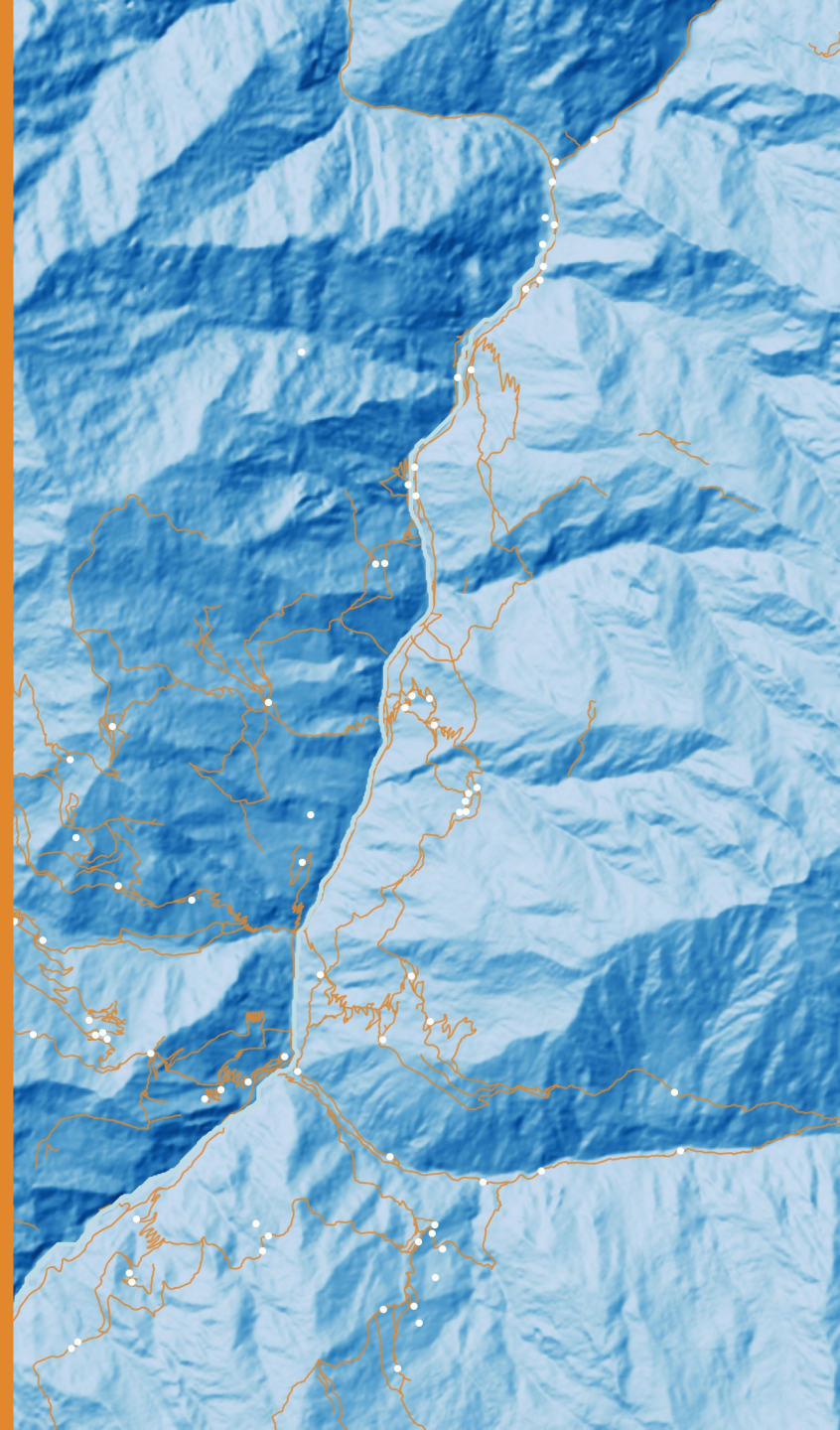
AI component: situation-report copilot, in English and Nepali

Water evidence: our UNet, trained from scratch on Kuro Siwo

Case study: Trishuli valley, Nepal, 26 August 2026

Team HimaRoad

Right: Trishuli valley relief from the Copernicus GLO-30 DEM, with OpenStreetMap roads and settlements from before the event.



THE PROBLEM

Rescuers need three answers, and they must trust every number

On 26 August 2026, a flood of water, mud and debris moved down the Bhote Koshi–Trishuli corridor in northern Nepal.

In such events, roads, bridges and phone lines can fail. Rescuers must know where the damage is and which villages are cut off.

Sentinel-1 radar sees through monsoon clouds. Sentinel-2 adds optical detail when the sky is clear.

1

Where did the flood hit?

The system compares Sentinel-1 radar images from before and after the event. It maps water and debris-like change.

2

What was damaged?

The system overlays OpenStreetMap buildings, roads and bridges from before the event. Overlaps are potentially exposed.

3

Who is cut off?

The system finds settlements that can lose the road connection to the nearest town or hospital.

THE COPILOT IS THE ANSWER INTERFACE

A rescuer asks a question in English or Nepali. The copilot answers from the saved maps of the system.

Each number in an answer or a report cites the evidence claim that it comes from.

Our UNet segmentation model and the radar change detector supply the water and change evidence.

EN + NE

answers and reports

Every number

cited to a claim

One page

PDF situation report

SYSTEM ARCHITECTURE

One pipeline from raw satellite data to cited answers

HIMAROAD MAPPING ENGINE

1 Allowed inputs

- Sentinel-1 GRD VV and VH
- Sentinel-2 L2A
- Copernicus GLO-30 DEM
- OSM before the event

2 Acquisition and pairing

- CDSE catalogue search
- Same-track pre/post pairs
- Gamma0 terrain correction
- 20 m UTM grid

3 Observability masks

- Layover and shadow from the DEM
- Provider data and shadow masks
- Unobservable means unknown

4 Hazard evidence

- **UNet water (our model)**
- Two-sided radar change and possible debris
- Sentinel-2 support
- 1,600 m² minimum unit

5 Exposure

- OSM roads, bridges and buildings
- Potentially exposed, observed with no change, or unobservable

6 Access scenarios

- Vehicle and pedestrian graphs
- Nearest town or hospital
- Four hypothetical scenarios

7 Persisted run bundle

- GeoJSON, rasters, report and manifest
- Producer commit, schema 3.0, scene IDs, thresholds and model version
- SHA256 checksum for every file

8 Situation-report copilot (AI component)

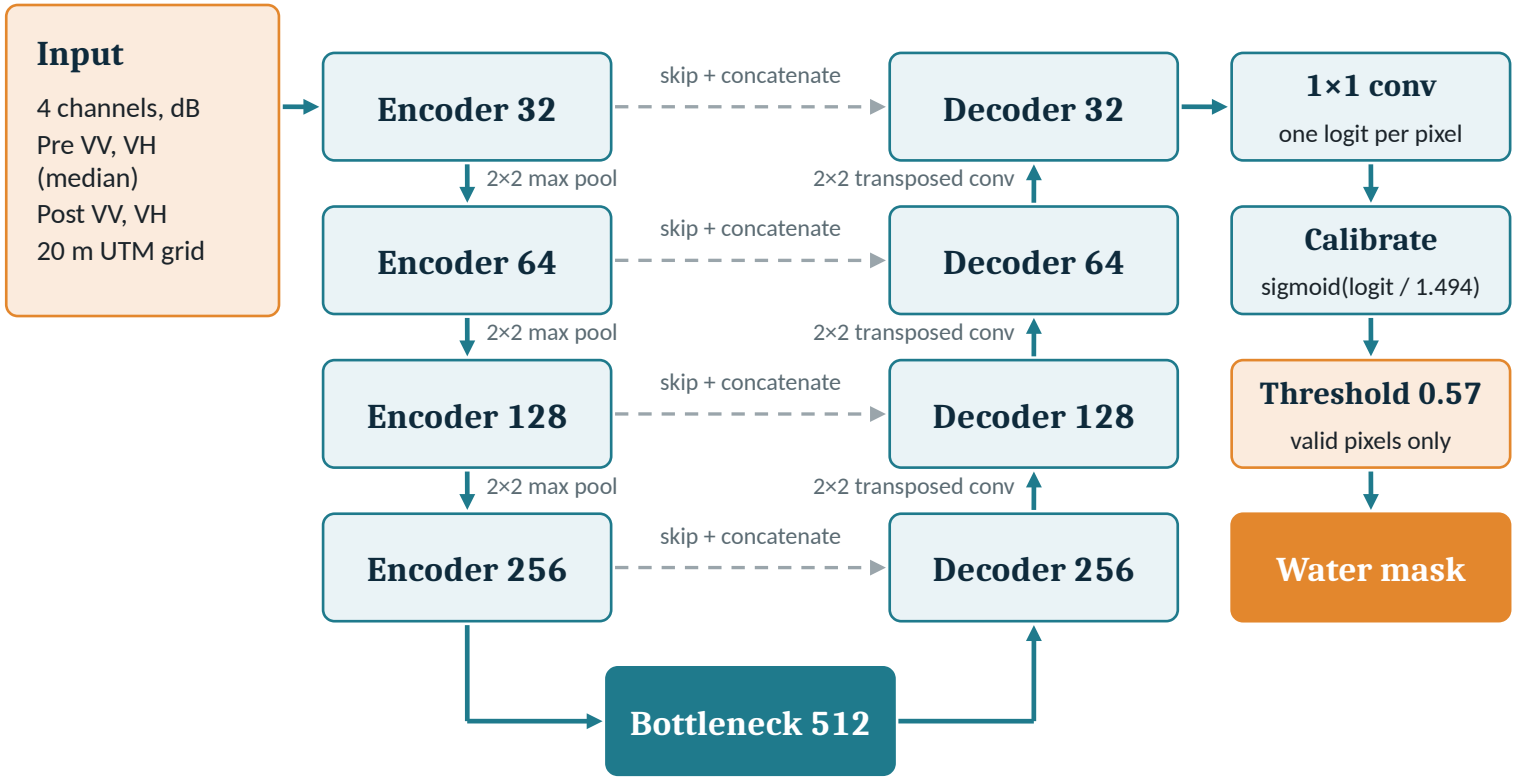
- Verifies the bundle against its SHA256 manifest
- Imports every number as an evidence claim
- Answers in English and Nepali, with a citation for each number
- Writes the one-page PDF situation report

9 Web app

- Map, copilot, evidence and report on one page
- English / Nepali switch
- Radar before / after slider
- **Bonus: DEM flow-path trace**

OUR SEGMENTATION MODEL

A UNet trained from scratch on Kuro Siwo maps water in radar pairs



Each encoder and decoder block: two 3×3 convolutions, each with GroupNorm and ReLU. Numbers give the channel width.

7,763,329
parameters

Depth 4
base width 32

From scratch
no pretrained encoder

Training and calibration

- Kuro Siwo labels, re-gridded onto CDSE gamma0 at 20 m.
- 310 training chips: Houston, Togo and Peru.
- 134 development chips: Larkana and Illinois.
- BCE + Dice loss, AdamW, learning rate 0.0003, batch 4, seed 42, 30 epochs.
- Four registered variants. The lowest development loss selected width 32 with 4 channels.
- Temperature 1.494 and threshold 0.57 were fixed on development data, at a precision floor of 0.80.
- Detector time for 600 km²: 1.7 s on GPU, 4.9 s on CPU (warm median).

FROM PIXELS TO KNOWLEDGE

The water mask becomes cited evidence for the copilot

1 UNet water mask
Calibrated probability ≥ 0.57 on valid pixels

→

2 Hazard policy
Removes permanent water and steep slopes
Keeps polygons of 1,600 m² or more

→

3 Hazard layers
UNet water-like change
Radar disturbance and possible debris
Optical support

→

4 Exposure and access
Each road link and building gets an evidence state
Graphs give four scenarios

→

5 Run bundle
SHA256 for every file
Model version and checkpoint SHA256 recorded

→

6 Evidence claims
One claim per number, with source file, field, method and hash

ONE REAL CLAIM FROM THE TRISHULI RUN

```
id        60ef6d96…18740a2445  (SHA256)
label     /summary/potentially_exposed_buildings
value     102   unit: count
scenario  2 (hypothetical)   status: unknown
source    report.json  /summary/potentially_exposed_buildings
sha256    15350b3c…a1903b75  (hash of report.json)
method    upstream-field   schema_version: 3.0
producer  commit 23a2527   flags include: hypothetical
```

WHAT THE COPILOT THEN SAYS

Potentially exposed buildings: 102 Hypothetical scenario: 2 [C60601]

सम्भावित जोखिममा रहेका भवनहरू: १०२ काल्पनिक परिदृश्य: 2 [C60601]

Lines from the English and Nepali situation reports. A bracketed reference resolves to the full claim ID in the export.

For a UNet run, model identity also becomes claims: model version, checkpoint SHA256, temperature and threshold.

THE COPILOT PIPELINE

Verify the run, cite every claim, then answer

PREPARE ONCE PER RUN

Run bundle
Map files, report.json and manifest.json

→

Hash check
SHA256 of every file must match the manifest

→

Evidence claims
Content-addressed claims in SQLite. Each has a source file, field, method and hash.

→

62,263 claims

Trishuli run, prepared in 44.2 s

EACH QUESTION

1 Question
One of 8 evidence questions, EN or NE

→

2 Refusal screen
Casualties, depth, collapse, safe routes, cause, forecasts

→

3 Fixed query
Selects the claims for this question from this run

→

4 Template answer
Authored EN or NE wording. Numbers come only from claims.

→

5 Session check
Every cited claim must belong to the loaded run

→

6 Cited answer
Numbered citations to SHA256 claim IDs

Fail → refusal with a reason. No number is shown.

ONE-PAGE REPORT

Same claims
The report uses the claims of the loaded run

→

Report facts
Area, satellite times, thresholds, coverage, assets, scenarios, limits

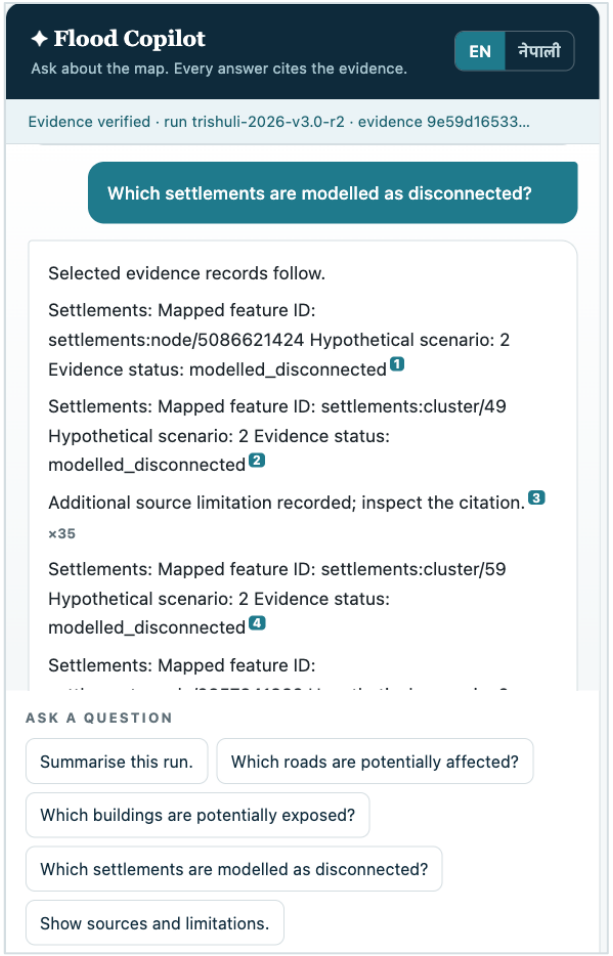
→

PDF, EN or NE
One A4 page. Nepali in Noto Sans Devanagari.

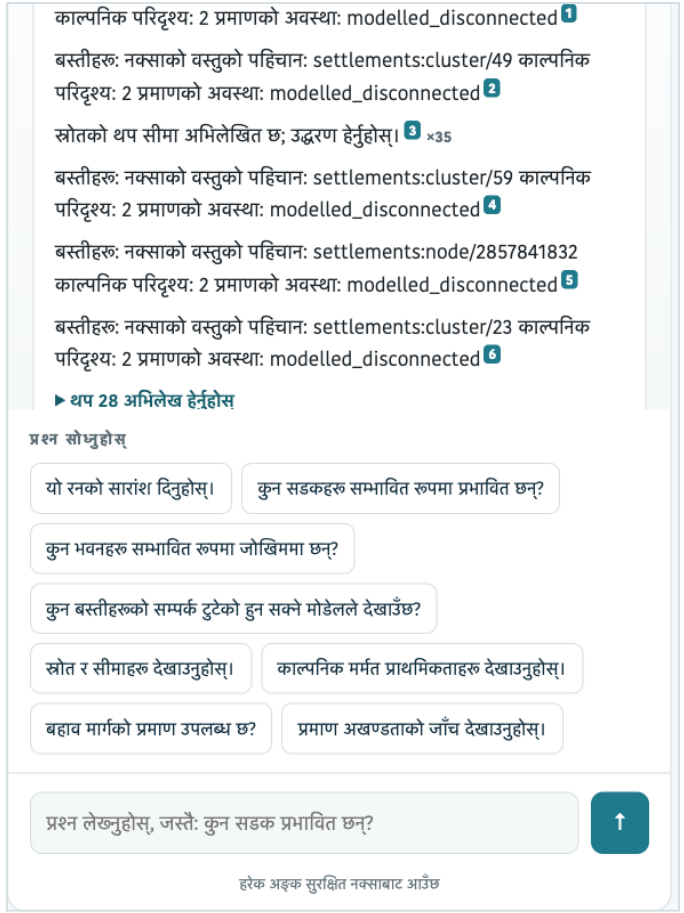
→

Exports
report-facts.json and report-claims.csv

Eight evidence questions, answered in English and Nepali



English: a cited answer with numbered citations, then the predefined questions.



Nepali: the same answer, the same citations and all eight questions.

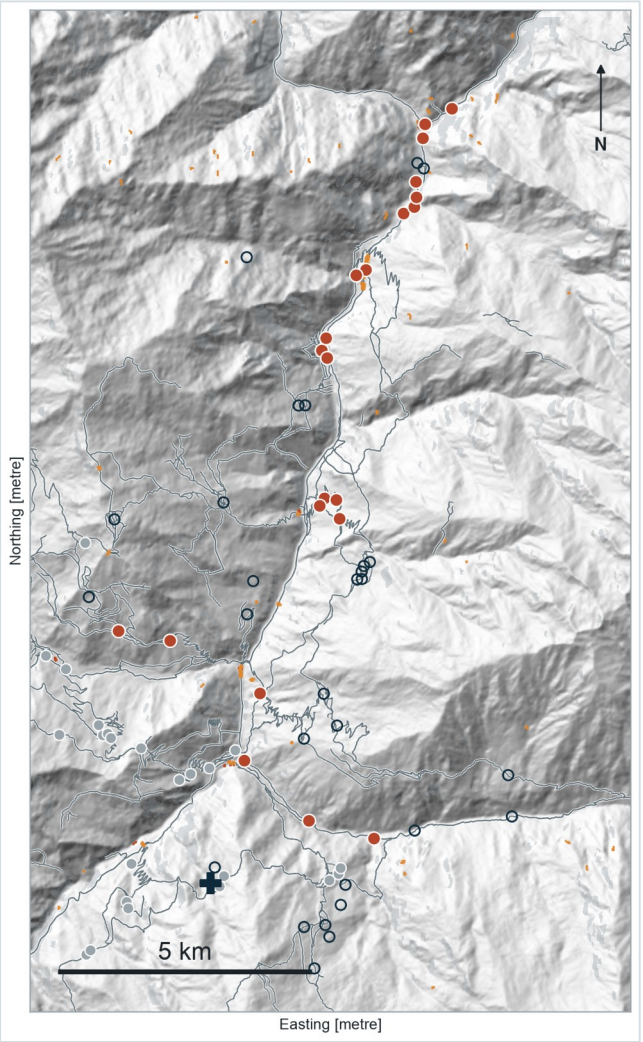
The copilot refuses, with a reason

- Casualty or missing-person questions
- Water depth and forecasts
- Confirmed collapse, destruction or closure
- Safe routes and the cause of the event
- Questions outside the loaded run

Checked before the submission

- 32 predefined cases (8 questions × EN/NE × 2 areas) passed validation, with exact claim-ID parity between English and Nepali.
- Nepali templates reviewed by a native Nepali speaker.
- A predefined answer takes 0.20–0.35 s on the Trishuli run.

Who may be cut off: 22 of 73 settlements in the primary scenario



MAP KEY (S2, PRIMARY)

- Potentially road-cut: 22
- Access uncertain: 23
- No mapped road access before the event: 28
- Town or health facility
- Suspected affected link
- Radar disturbance
- Possible debris (unvalidated)
- Unobservable (unknown)

Trishuli run on a Copernicus DEM hillshade. Scale bar: 5 km.

Hypothetical scenario	Links removed	Road-cut
S1	Independently sourced confirmed closures	0
S2 (primary)	S1 plus links with observed change or likely impact	**22 of 73**
S3	S1 plus radar-optical corroborated links	0
S4 (worst case)	S2 plus possible-impact, unobservable and unknown links	45 of 73

14 of 22
potentially road-cut settlements have a mapped trail

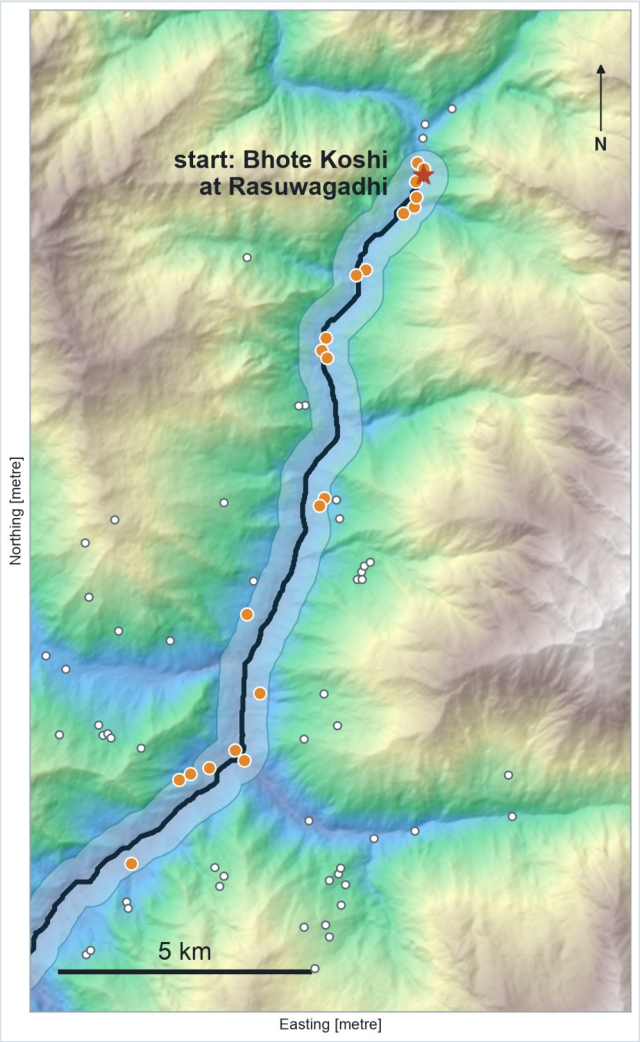
0
primary-care disconnections in S2

47 of 73
settlements are clusters of OSM buildings

- Directed vehicle and pedestrian graphs from OSM before the event.
- Access before the event is checked first. A settlement without a prior connection is not a new cut-off.
- Each settlement gets its nearest reachable town or health facility and a candidate route.
- All scenarios are hypothetical. A route is mapped reachability, not a safe route.

BONUS: DEM FLOW PATH

Trace the flow path down the valley from any upstream point



27.0 km

D8 flow path from the Bhote Koshi at Rasuwagadhi to the edge of the grid

21 settlements

within 500 m of the path, listed in downstream order

- Condition the Copernicus GLO-30 DEM with priority-flood filling.
- Follow D8 steepest descent from the start point.
- List the settlements within 500 m of the path, in downstream order.
- The trace shows where water from the point goes on the DEM surface. It is a terrain trace, not an inundation forecast.

★ Start point — Flow path

● In the 500 m corridor — ○ Other settlement

Flow path from an upstream point

Click the map to pick a start point, then trace the path down the valley on the Copernicus DEM. This is a terrain trace, not an inundation forecast.

Trace flow path Start: 28.2710° N, 85.3780° E

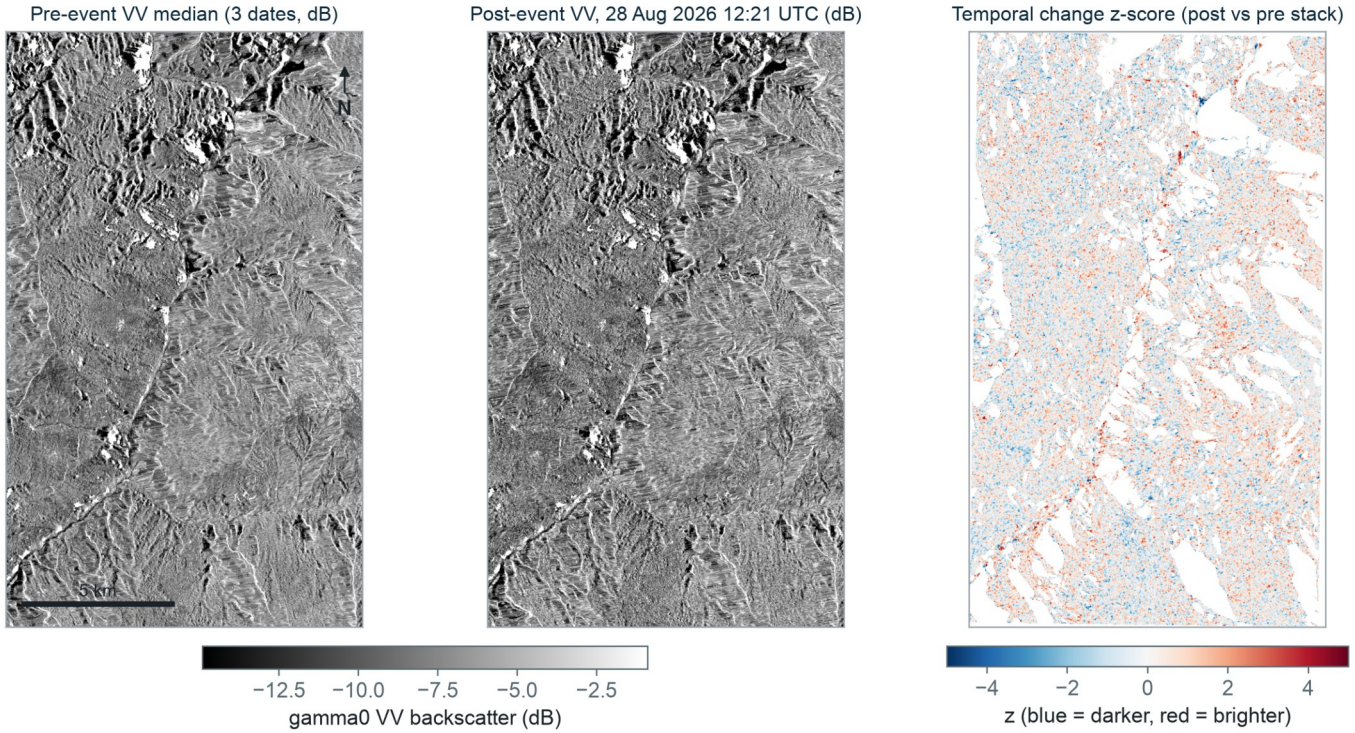
27.0 km downstream, stops at grid_edge. 21 settlements within 500 m, in downstream order:

1. Building cluster 55
2. Building cluster 57
3. Building cluster 53
4. Building cluster 54
5. Building cluster 52
6. Building cluster 51
7. Building cluster 49
8. Building cluster 44

Web app: click the map, then click "Trace flow path". The engine traces the path live on the run's DEM.

CASE STUDY

Trishuli, 26 August 2026: the frozen run



Sentinel-1 VV gamma0: pre-event median of three dates, post-event 28 Aug 2026 12:21 UTC, and the temporal change z-score.

102

potentially exposed buildings (overlap with a hazard, not damage)

121

suspected affected road links (operational status unknown)

22 of 73

settlements potentially road-cut (S2, hypothetical)

93.97%

valid change observation

99.08%

valid post-event observation

24.9%

of the area masked by layover or shadow (unknown, not zero)

Area 85.30–85.42° E, 28.12–28.30° N. **Radar after the event** 28 Aug 2026 12:21 UTC, orbit 85 ascending, 60.4 h after the event-date boundary. **Radar before** 23 Jul, 4 Aug and 16 Aug 2026, same track. **OSM snapshot** 2026-08-25 00:00 UTC. **Provenance** producer commit 23a2527, schema 3.0, frozen (tag case-study-frozen) before EMSR927 was opened.

SPECIAL CONSIDERATIONS

Rules the system keeps in every run and every answer

1 **Data firewall**

EMSR927, UNOSAT and GFM are for checking only, after the freeze. They never enter inference, thresholds, training or model selection.

2 **OSM from before the event**

The snapshot is at 00:00 UTC on the day before the event. Each run records an edit-timestamp proof.

3 **Same-track pairing**

Only scenes from the same orbit track are compared. With no pair, the run ends as incomplete, with the reason.

4 **Layover and shadow masks**

Steep slopes that the radar cannot see are unobservable. Unobservable means unknown, never zero.

5 **Careful wording**

Overlap with a hazard means potentially exposed. Operational status stays unknown. Scenarios are hypothetical.

6 **Satellite time is not event time**

Each output records the scene time and its offset from the event-date boundary.

7 **Provenance in every output**

Producer commit, schema version, scene IDs, thresholds and model version travel with each run.

8 **Offline run bundles**

SHA256 checks on every file. A changed file stops the import. Saved runs work without a network.

9 **Bilingual output**

English and Nepali answers share the same claims. Nepali templates reviewed by a native Nepali speaker.

10 **Every number cited**

Each number in an answer or report cites a claim from the maps. No casualty figures, no event cause, no safe routes.

ACCURACY

What we measured, including the weak results

1 UNet on development data

Method	Precision	Recall	IoU	F1
Rule baseline	0.8835	0.6629	0.6096	0.7575
UNet	**0.8008**	**0.9457**	**0.7656**	**0.8672**

1,167,702 valid pixels from Larkana and Illinois.

These events selected the model. Thus, these are development numbers, not independent accuracy.

2 Blind Himalayan test, once only

0 of 1,298

labelled new-water pixels found by the UNet (recall 0, IoU 0). The rule baseline found 24 (precision 0.828, recall 0.018).

Teesta 2023 (81 px), Melamchi 2021 (970 px) and Chamoli 2021 (247 px). Two independent reviewers labelled blind windows.

The model does not yet transfer to narrow Himalayan valleys. We changed no threshold, model or checkpoint after the test.

3 EMSR927 check, after the freeze

AOI	Landslide area recall	Affected transport on exposed links
AOI01 Syapru Besi	**0.012**	**6 of 19**
AOI02 Timure	**0.038**	**7 of 13**

Water IoU is unavailable: the reference grades landslides only.

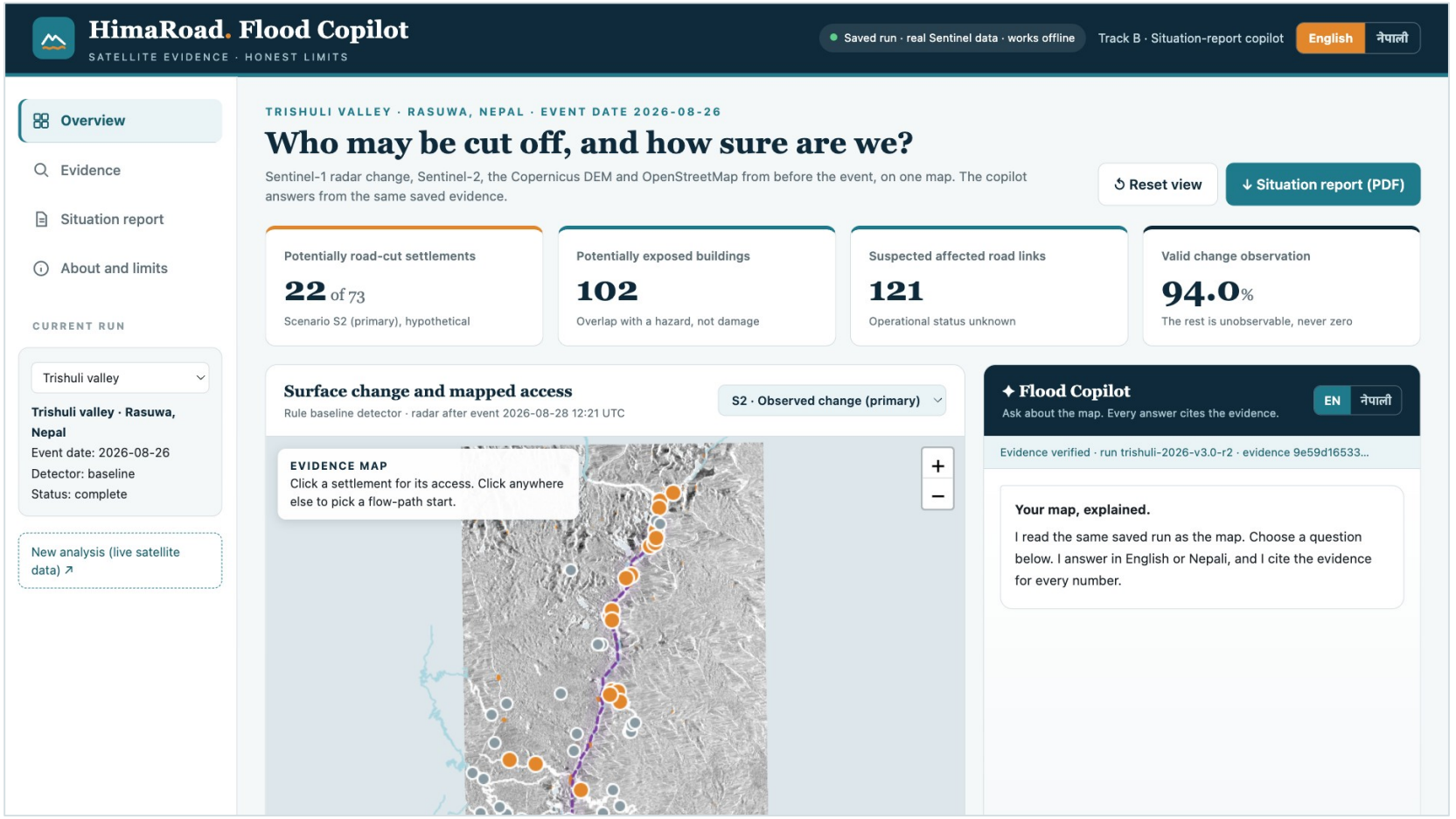
0 of 8 "no visible damage" features flagged in AOI01.

We observed 93.1% and 76.1% of the landslide footprints. The low recall is a detection miss, not a visibility gap.

In the case study, the UNet run and the rule run are identical on every scored item. The minimum-mapping-unit and permanent-water step removed the one UNet water candidate. Thus, the compared disturbance layer is rule evidence, not model output.

THE WEB APP

Map, copilot and evidence on one page



Overview page of the saved Trishuli run, at 1920 × 1080.

1 Metric cards

Saved results of the run. The road-cut count changes with the scenario.

2 Scenario menu

Selects one of the four hypothetical scenarios.

3 Evidence map

Radar image, hazard layers, roads by evidence class, settlements. A slider compares radar before and after.

4 Copilot panel

Predefined questions in EN or NE, with numbered citations.

5 Other pages

Evidence (scenes, gaps, OSM snapshot, thresholds), Situation report, About and limits.

SITUATION REPORT

A one-page report in English and Nepali, from the same claims



Situation report page of the web app.

Situation report

Potentially affected mapped assets

Settlements with uncertain access: 23 Hypothetical scenario: 2 **[C60598]**
Potentially exposed buildings: 102 Hypothetical scenario: 2 **[C60601]**
Modelled disconnected settlements: 22 Hypothetical scenario: 2 **[C60602]**
Potential primary-care disconnections: 0 Hypothetical scenario: 2 **[C60603]**
Potentially affected links: 121 Hypothetical scenario: 2 **[C60606]**
Possible mapped trail access: 14 Hypothetical scenario: 2 **[C60607]**

English PDF, Trishuli run: title and the section "Potentially affected mapped assets".

अवस्था प्रतिवेदन

सम्भावित प्रभावित नक्साङ्कित सम्पत्ति

पहुँच अनिश्चित भएका बस्तीहरू: २३ काल्पनिक परिदृश्य: 2 **[C60598]**
सम्भावित जोखिममा रहेका भवनहरू: १०२ काल्पनिक परिदृश्य: 2 **[C60601]**
मोडेलमा सम्पर्क टुटेका बस्तीहरू: २२ काल्पनिक परिदृश्य: 2 **[C60602]**
स्वास्थ्य केन्द्रसँग सम्भावित सम्पर्क विच्छेद: ० काल्पनिक परिदृश्य: 2 **[C60603]**
सम्भावित प्रभावित सडक खण्डहरू: १२१ काल्पनिक परिदृश्य: 2 **[C60606]**
नक्सामा सम्भावित गोरेटो पहुँच: १४ काल्पनिक परिदृश्य: 2 **[C60607]**

Nepali PDF: the same section, with Devanagari numerals and the same claim references.

- The copilot writes the report from the claims that the answers use.
- Each line ends with a claim reference. The exports resolve each reference to the full claim ID.
- Sections: area and event, satellite times, thresholds, coverage, hazard areas, affected assets, scenarios, limitations and credits.

Data attributions and training-dataset citations

Sentinel-1 and Sentinel-2	“Contains modified Copernicus Sentinel data 2026.”
Copernicus DEM	“Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved.”
	“The organisations in charge of the Copernicus programme by law or by delegation do not incur any liability for any use of the Copernicus WorldDEM-30.”
OpenStreetMap	“© OpenStreetMap contributors.” Data from before the event, under the ODbL.
Kuro Siwo	Bountos et al., NeurIPS Datasets and Benchmarks 2024 (training data of the UNet).
Sen1Floods11	Bonafilia et al., CVPR Workshops 2020 (surveyed, not used).
Copernicus EMS	“European Union, Copernicus Emergency Management Service data” (EMSR927, checking only).

Code: Apache-2.0. Web map: Leaflet 1.9.4 (BSD-2-Clause). Nepali font: Noto Sans Devanagari (SIL Open Font License). Data, model weights and reference products keep their own terms.